Fine-Tuning Language Models as Multimodal Differential Equation Solvers

ICON-LM — Yang, Liu, & Osher (2024)

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Why ICON-LM?

Classic Operator Learning

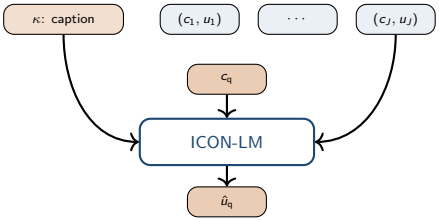
- Trains 1 neural operator for 1 map $T : c \mapsto u$
- Change the PDE, coefficients, or task $\Rightarrow$ Retrain or fine-tune
- Hard to inject equation knowledge without redesigning the model

ICON-LM

- Treat examples $\{(c_j, u_j)\}$ as the *context*
- Treat a text/equation caption κ as another input modality
- Predict the query function in one forward pass with no weight updates

Changes introduced by ICON-LM

1. **Multimodal prompts**: Data examples with a caption to describe the operator
2. **LM-style training**: 1 transformer encoder trained by next-*function* prediction



Paper benchmark: 19 ODE/PDE/control operator families.

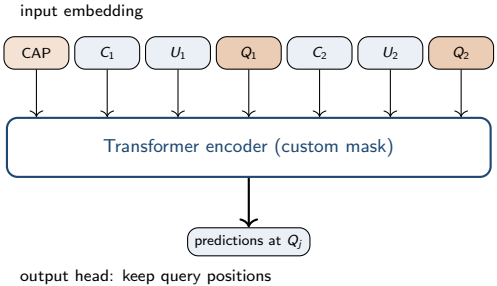
Architecture

From encoder–decoder ICON to ICON-LM

- Original ICON: two transformers ($\sim$ 31.6M params).
- ICON-LM: a single GPT-style encoder ($\approx$ 15.8M).

Tokens:

- Three token types per example: **C**ondition, **U** (QoI), **Q**uery.
- Query tokens carry spatial *locations*; values zeroed out (model fills these in).



The Custom Mask: Preventing Leakage

Why it is needed

- Each example has an input function C_j and output function U_j
- To predict U_j, the query tokens Q_j may use previous example pairs alongside C_j
- But Q_j must not attend to the current output U_j, because that is the answer
- Previous queries unused as context

Key idea

- White blocks are allowed context, grey blocks are blocked to prevent answer leakage.

Q_j can see $(C_1, U_1), \ldots, (C_{j-1}, U_{j-1}), C_j$

Q_j cannot see U_j

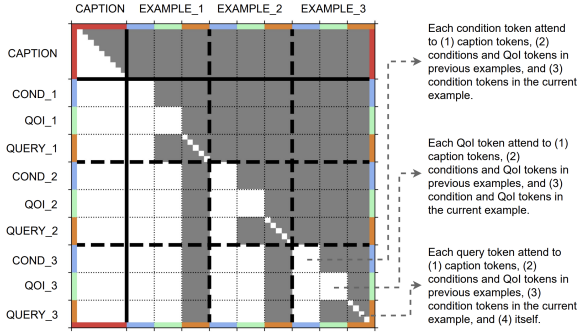


Figure 3 from Yang, Liu, & Osher

Training

Autoregressive loss across examples

For each $J = 0, 1, \ldots, I-1$ in an I-example prompt, predict u_{J+1} from earlier examples and the caption:

$$\hat{u}_{J+1} \;=\; T_\theta\big[c_{J+1};\, \{(c_i, u_i)\}_{i=1}^{J},\, \kappa\big]\,.$$

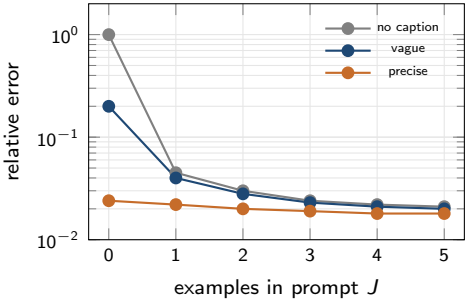
$$\mathcal{L}_{\text{prompt}} \;=\; \sum_{J=0}^{I-1} \big\|\hat{u}_{J+1} - u_{J+1}\big\|^2.$$

- One prompt simultaneously trains *every* few-shot count J.
- With a caption, $J = 0$ contributes (zero-shot); without, training starts at $J = 1$.
- Multi-modal loss = loss with caption + loss without caption.
- **Inference**: append a input condition c_{q} + query tokens, one forward pass.

Paper Results

Main empirical claims

- **Smaller and stronger**: ICON-LM improves on encoder–decoder ICON with roughly half the parameters
- **Captions matter**: Precise captions enable zero-shot prediction; vague captions reduce the number of examples needed



Model	Params	Use at test time
Encoder–decoder ICON	$\sim$ 31.6M	examples only
ICON-LM	$\sim$ 15.8M	examples + caption

Caption benefit is most visible at $J = 0$ or $J = 1$, exactly where numerical examples are sparse.

Ablation

Zero-shot relative error with precise captions

GPT-2 init.	Train	Test	Gap
Pre-trained weights	2.30 %	2.44 %	**0.14 %**
Random weights	2.07 %	3.08 %	1.01 %

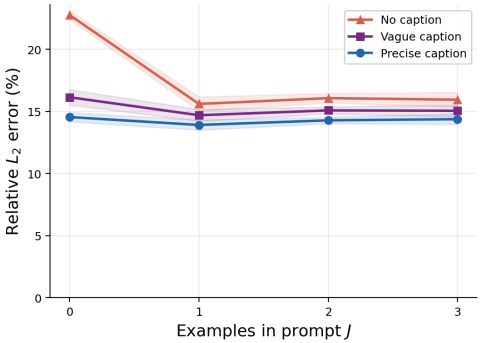
- **Pre-trained** GPT-2 generalizes better across captions
  - Suggests language priors transfer
- **Random init** can win in few-shot settings
  - Suggests language pre-training is not always ideal for numerical prediction

Reproduction on Burgers' Equation

What we reproduced

$$u_t + u\,u_x = \nu\,u_{xx}$$

- Reproduced ICON-LM architecture: custom mask + next-function loss
- Applied to 1D viscous Burgers' equation, operator family indexed by ν.
- 2.2M-parameter transformer; 5,000 gradient steps.
- Optional captions via frozen MiniLM embeddings.



Conclusion

Key Takeaway

ICON-LM turns operator learning into a language-model-style problem: examples are context, functions are tokens, captions are side information, and a custom mask makes next-function prediction possible without leakage.

Why it works

- Single transformer encoder.
- One training sequence supervises all few-shot levels at once.
- Captions help when examples are scarce.

Thanks for listening!

References

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- Code: `github.com/LiuYangMage/in-context-operator-networks`